Preface

In the fields of data analysis and financial security, fraud detection has emerged as a crucial and difficult job. The necessity for efficient techniques to recognize and stop fraudulent activity has increased due to the number and complexity of digital transactions. Advanced machine learning approaches are required because traditional rule-based systems are sometimes unable to tackle the complex strategies used by fraudsters.

The fascinating subject of fraud detection with auto-encoders—a kind of neural network that is excellent for anomaly detection and unsupervised learning—is explored in this research. Because auto-encoders are capable of both compression and reconstruction, they are perfect for spotting anomalies like fraudulent transactions. We seek to identify transactions that considerably depart from the norm, suggesting possible fraud, by concentrating on the reconstruction error.

In this study, I investigate the creation and assessment of an auto-encoder model intended especially for transaction data fraud detection. The goal is to build a resilient system that does not require large amounts of labeled data in order to respond to novel and changing fraud trends. This strategy is especially helpful in the banking industry since fraud techniques are always evolving.

I want to learn more about the potential and constraints of auto-encoders for fraud detection by carefully examining the model's performance and efficacy using pertinent evaluation measures, such as reconstruction error levels and visual inspection. These results will support continuing efforts to improve fraud detection systems and lay the groundwork for future studies examining the use of machine learning methods to financial security.